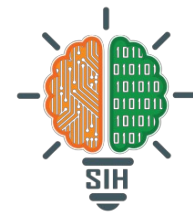


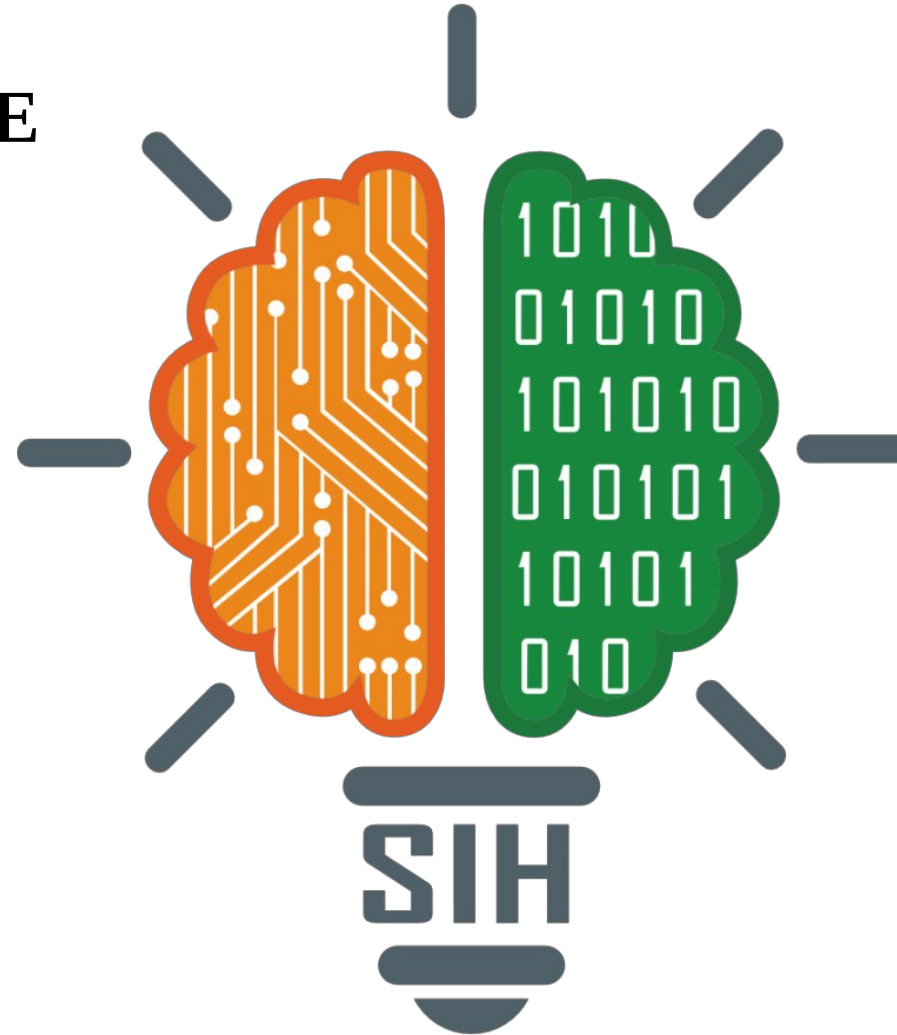
SMART INDIA HACKATHON 2026

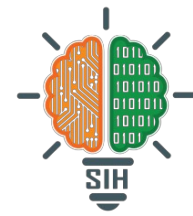


SMART INDIA
HACKATHON
2026

TITLE PAGE

- Problem Statement ID – _____
- Problem Statement Title- _____
- Theme- **Artificial Intelligence (AI)**
- PS Category- **Software**
- Team ID- _____
- Team Name (Registered on portal)- **Sankalp**





❖ Proposed Solution

• Detailed explanation of the proposed solution

- Keeps ONE small AI engine in memory – not ten heavy ones
- Snaps on a tiny “skill cap” for the job: coding, maths, writing
- Swaps caps live – no restart, no memory flush, no freeze
- Runs as a silent local server on a loopback address

• How it addresses the problem

Token Tax

→ Everyday tasks run free, on your own PC

GPU Memory Wall

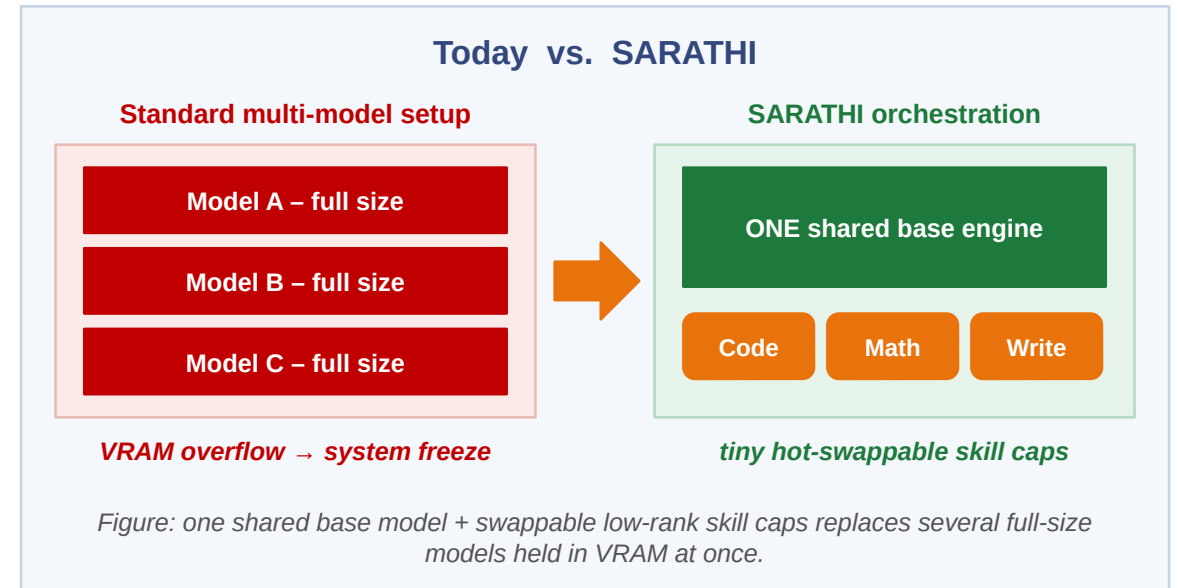
→ One base engine + featherweight skill caps

Context Loss

→ Hybrid memory re-injects history on every swap

• Innovation and uniqueness of the solution

- Auto hardware profiler picks the right-sized free model for YOUR PC
- Live cap-swapping without flushing GPU memory – context survives
- Speaks MCP, so it plugs into the tools you already run



Lower running cost

Routine work stays on your own PC instead of a paid cloud API

Fits ordinary hardware

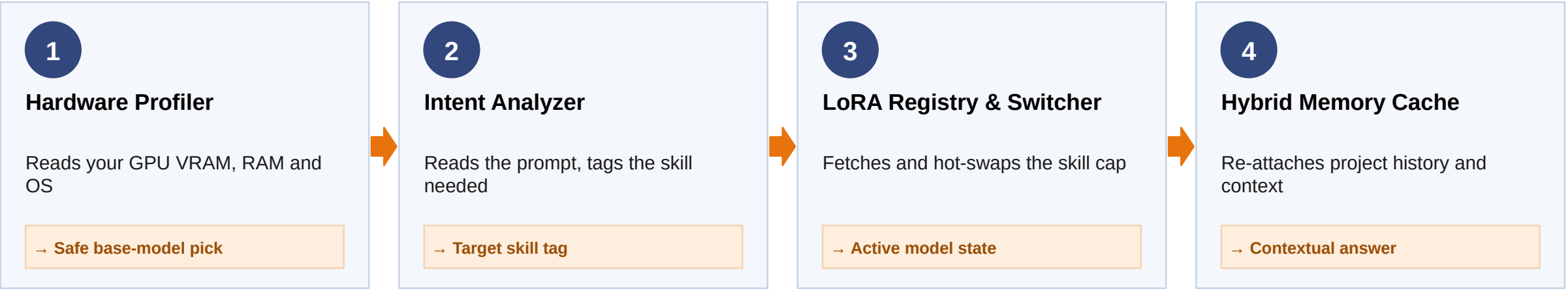
One base engine held in memory instead of several full models

Private by default

Prompts and files are processed on your device

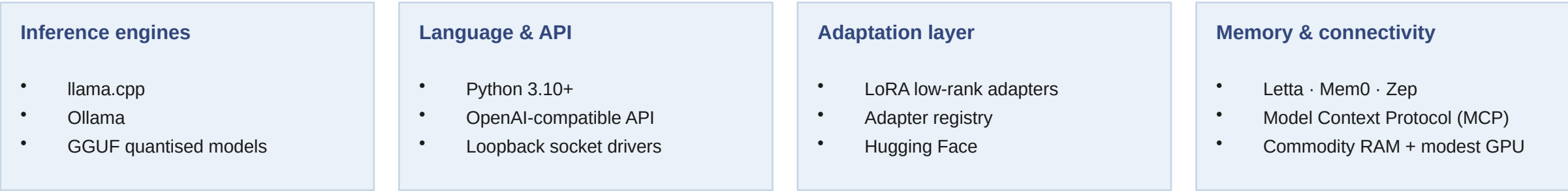
TECHNICAL APPROACH

• **Methodology and process for implementation (flow of every prompt)**



Runs silently in the background as a local server on a loopback address – nothing leaves the machine during use.

• **Technologies to be used (languages, frameworks, hardware)**



Adapter fine-tuning is performed externally – SARATHI orchestrates and serves.

• Analysis of the feasibility of the idea

Working proof of concept

✓ A compact base model (Qwen3 4B) already orchestrated with externally fine-tuned adapters.

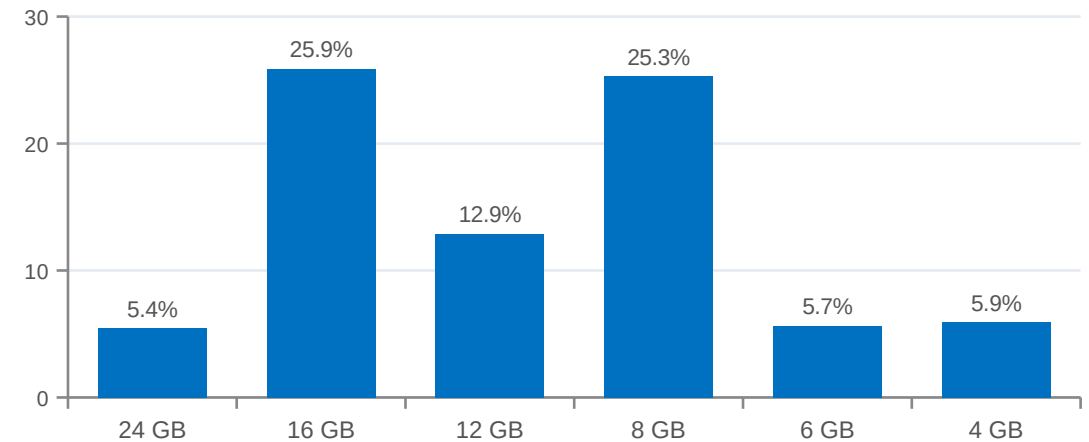
Mature open-source base

✓ llama.cpp and Ollama already support request-level matrix modification.

Modular team split

✓ Profiler, Classifier, Memory Cache and Adapter Registry build and test independently.

Consumer GPU VRAM in the wild (Steam, Jul 2026)



Half of all gaming PCs now hold 10 GB+ VRAM – still not enough for several full models at once. That gap is SARATHI's market.

• Potential challenges and risks → Strategies for overcoming these challenges

RISK

A skill cap fails to load

An adapter may be missing, corrupt or incompatible with the running base engine.



MITIGATION STRATEGY

- Runtime validation checks the adapter before it is applied
- Error boundaries contain the failure instead of crashing the app
- Graceful fallback: the base engine answers while the cap is skipped

• Potential impact on the target audience

Students & solo developers

Full AI tooling with no monthly bill

Small firms & startups

Client code and data never leave the office

Regulated & rural users

Works offline; nothing crosses a border

• Benefits of the solution (social, economic, environmental)

What changes for the user

Cloud AI today	With SARATHI
Pay a fee for every task	Routine tasks run on hardware you own
Prompts and files sent to a server	Prompts and files stay on the device
Needs a live internet connection	Keeps working offline
The vendor decides the model	You choose the model and the skills

Every routine task kept off the cloud is a recurring bill that never starts.

Social	Economic	Environmental
- AI access without a credit card or subscription - Works where the internet does not – offline by design - Personal and medical data stay on the user's device	- Removes per-token fees for everyday tasks - Reuses hardware students and firms already own - Skill files are small – cheap to store and share	- Small local model instead of a data-centre call for simple work - Less GPU memory per task means less power drawn - Extends the useful life of existing PCs

✓ No per-token bill

✓ Runs on hardware you own

✓ Data stays on your device

✓ Works without internet

• Details / Links of the reference and research work

1

Hu et al., “LoRA: Low-Rank Adaptation of Large Language Models”, ICLR 2022

Concept used in SARATHI: Freeze the base model and train only small low-rank matrices – this is exactly what a SARATHI “skill cap” is.

arxiv.org/abs/2106.09685

2

Sheng et al., “S-LoRA: Serving Thousands of Concurrent LoRA Adapters”, MLSys 2024

Concept used in SARATHI: Keep adapters in system RAM or on disk and page them into GPU memory just-in-time – the design of our LoRA Registry & Switcher.

arxiv.org/abs/2311.03285

3

Addair et al., “LoRAX: Multi-LoRA Inference Server for Production Workloads”, 2024

Concept used in SARATHI: Asynchronously fetch an adapter from a registry (Hugging Face or a local folder) and exchange it between GPU and CPU memory without restarting the engine.

github.com/predibase/lorax

4

Gerganov et al., “llama.cpp: C/C++ LLM Inference Engine”, 2024

Concept used in SARATHI: The quantised GGUF runtime SARATHI actually runs on – dependency-free local execution on an ordinary PC.

github.com/ggml-org/llama.cpp

How the research fits together

- [1] gives us the small, swappable skill file
- [2] and [3] give us the way to load and unload it without restarting
- [4] gives us the engine that runs it on a normal PC
- SARATHI's own contribution: bringing these together on a single-user desktop, with memory that survives the swap

Standards & ecosystem

- Model Context Protocol – open tool-connection standard, now under the Linux Foundation
- Ollama, Hugging Face, Letta, Mem0, Zep – provider ecosystem

Guided by Prof. Rupali Zambre

Pimpri Chinchwad University, Pune – CSE-AIML

Prototype validated on a compact base model with externally fine-tuned adapters.